Basics on Probability

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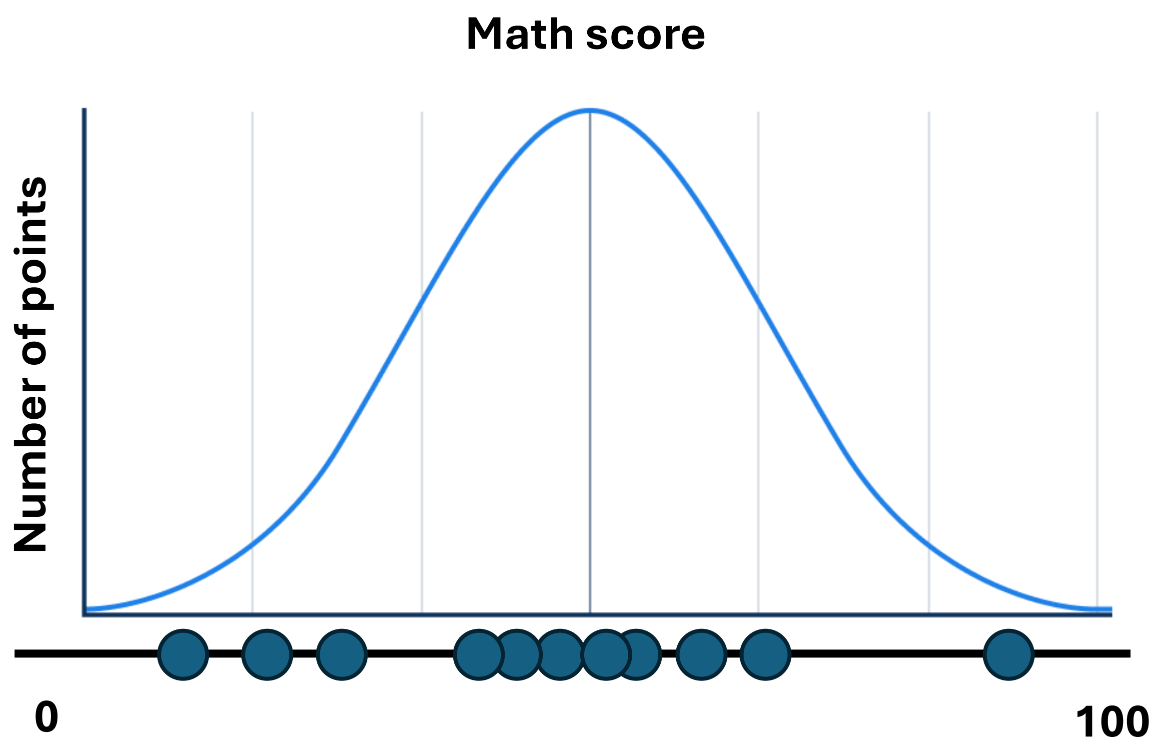
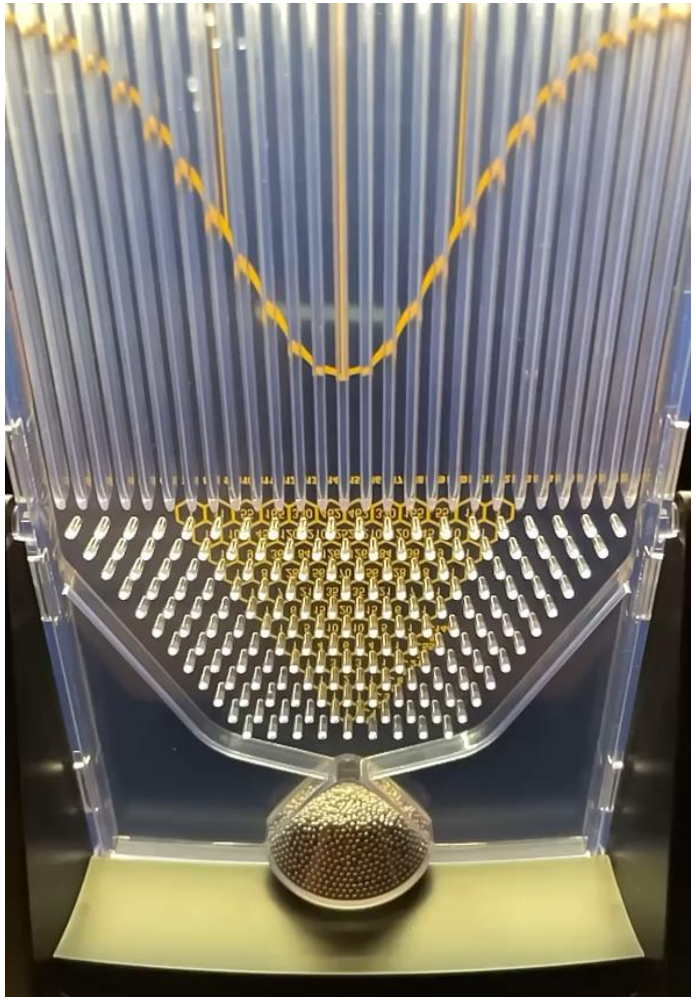
- Many contents are inspired from CS 285 at UC Berkeley
  https://www.youtube.com/@rail7462/courses
- Check https://github.com/bear-research-lab/Class_AIR/tree/main/Week3 for extra codes.

B0. Probability density function



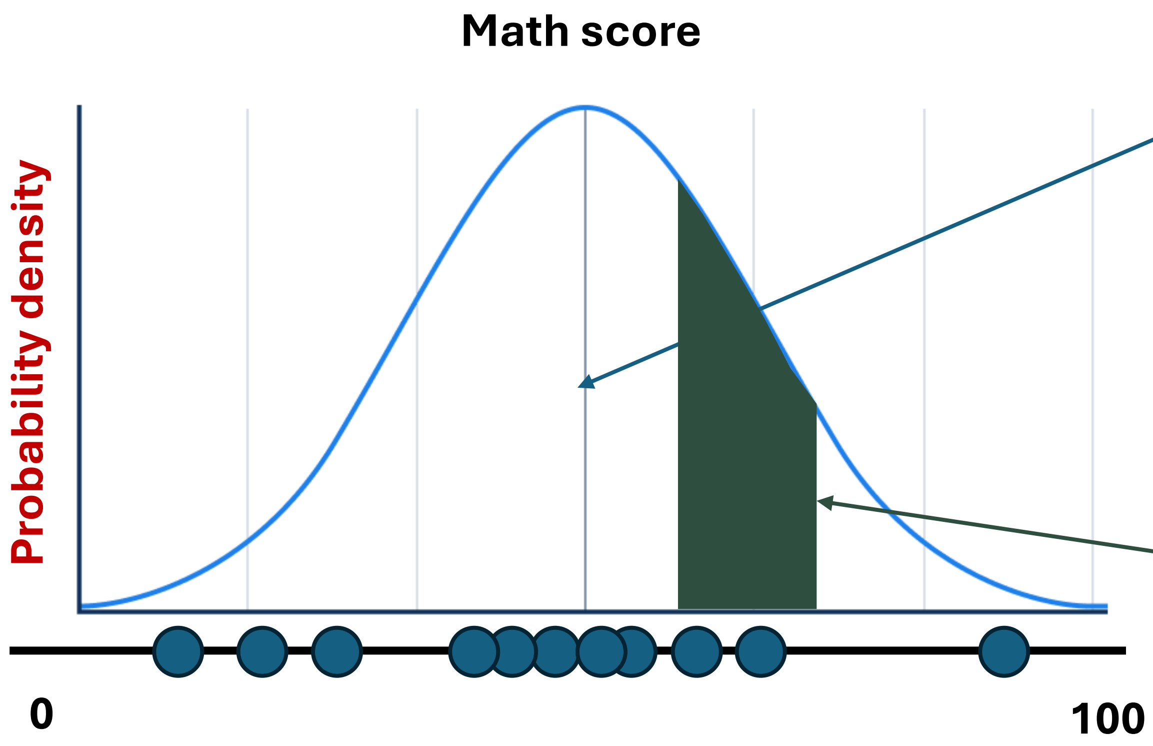


- Y axis means how many data will be there in this phenomena.

https://youtube.com/shorts/TwctT3Ncm1w?si=RdZ4ZW7auyRFpCWs

Basics of Probability and Machine Learning

B0. Probability density function – Definition and usage



1. Measure the whole area beneath the curve

2. Rescale the graph by dividing the y value by measured area. [$P(x) = f(x) / \int f(x)dx$]

3. Now the area beneath the curve will be 1.

4. What does this area size mean?

Probability of how the math score will be between a~b.

- The function that represents this graph is called: “probability density function (PDF)”.
- The area beneath the curve defines the probability.

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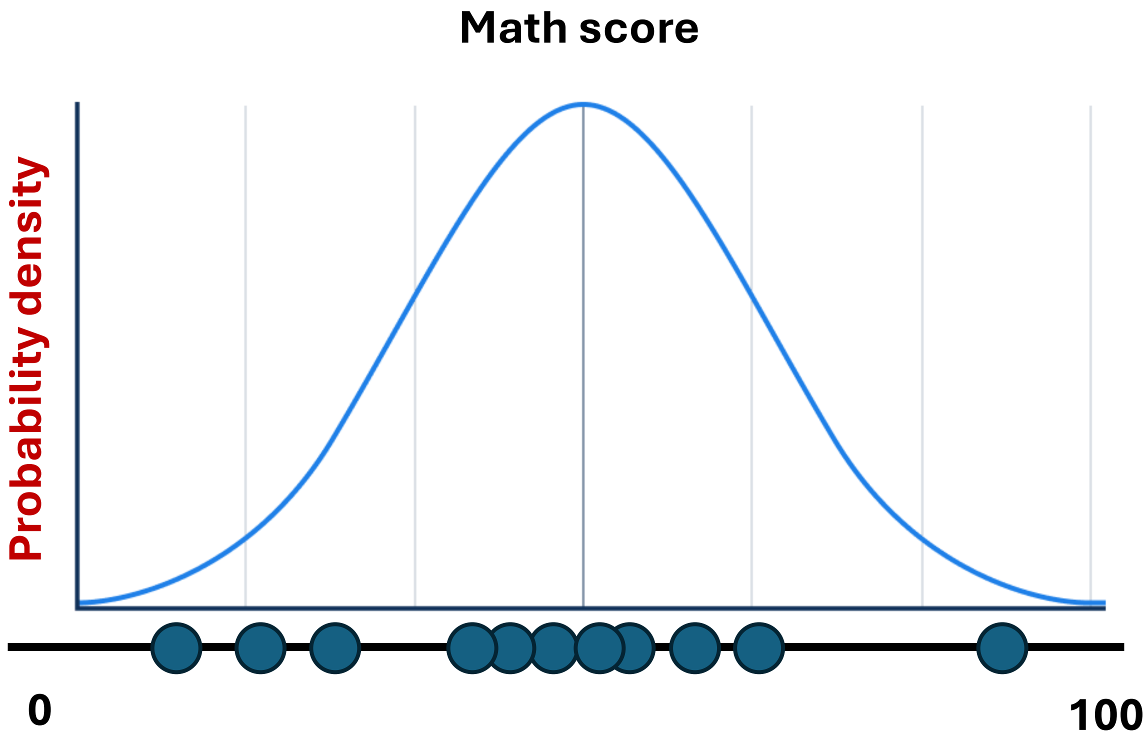
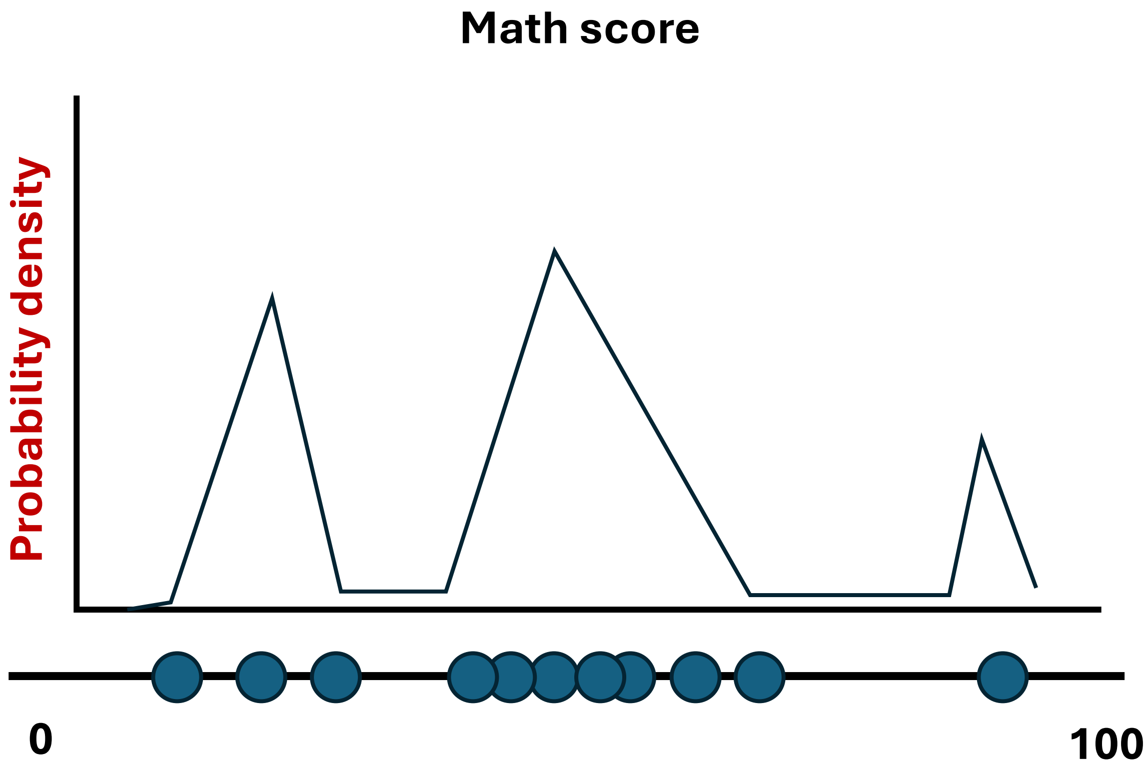
B0. Probability density function





- Which distribution (shape) is better to represent our math score data?

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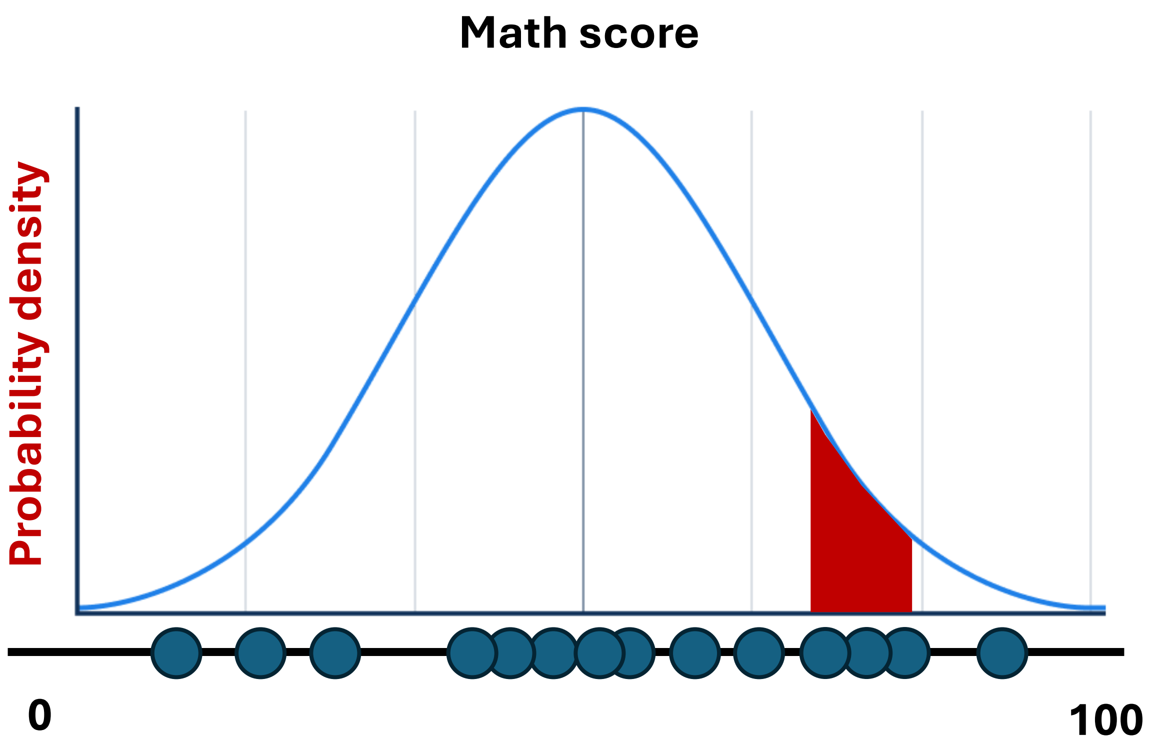
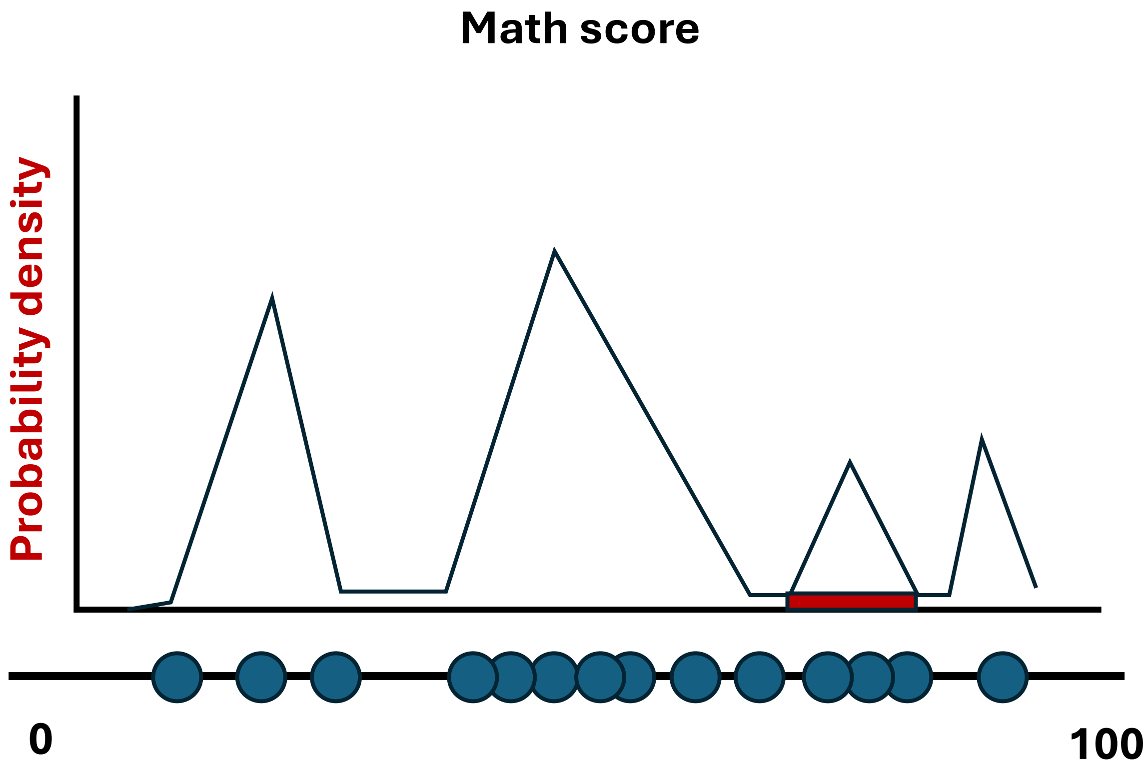
B0. Probability density function





- What is the probability of students to get score between 75~90?
- The left distribution assigns higher probability mass to the observed data range.

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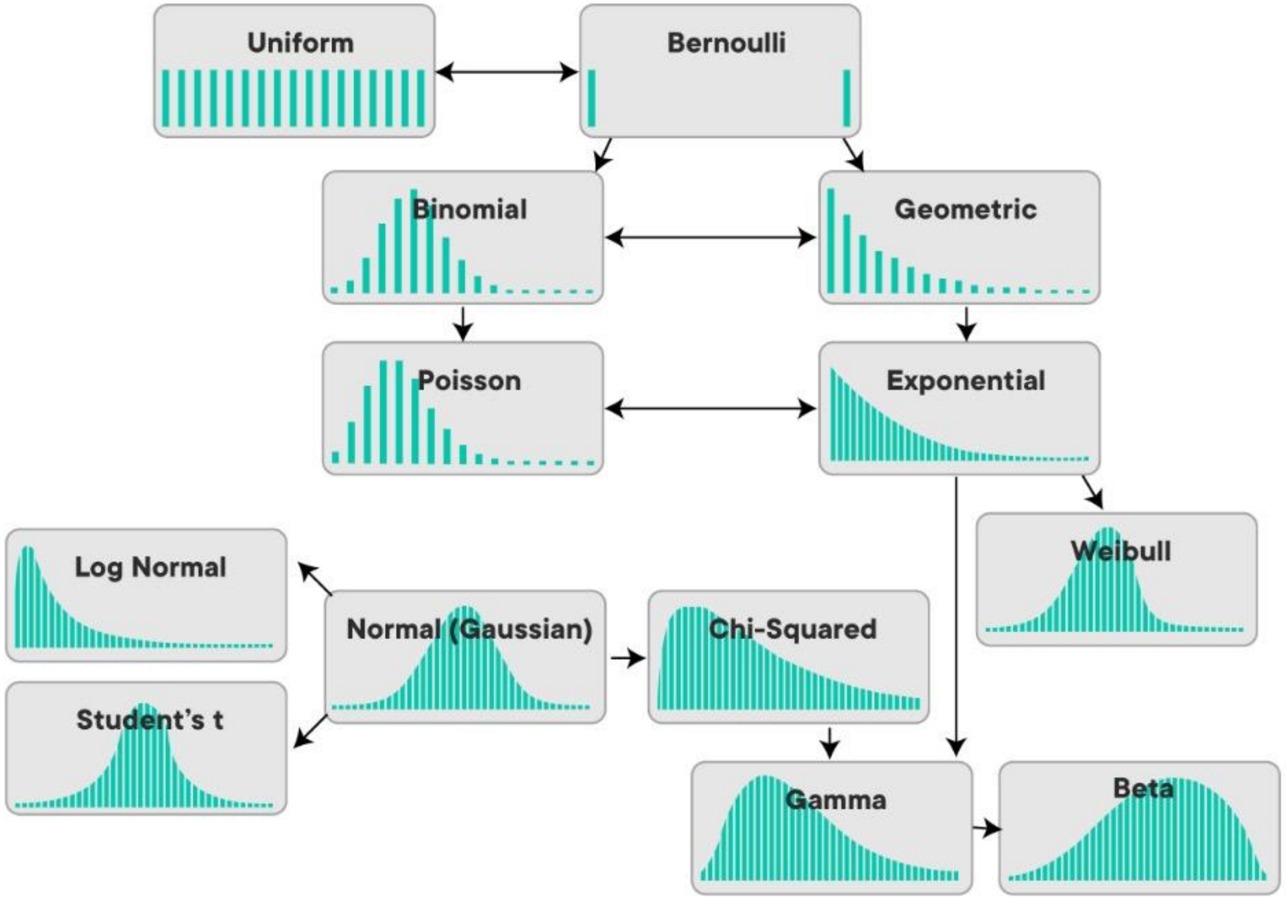
B0. Probability density function – We need to choose an "appropriate" distribution

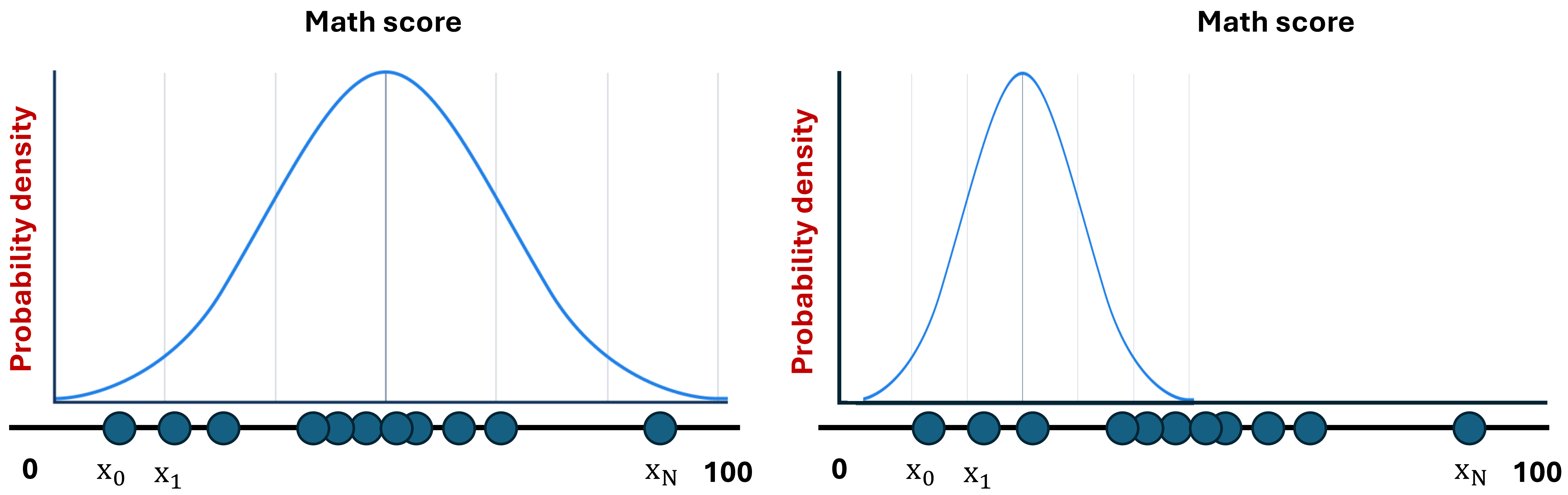


- There exists diverse distributions that can represent Probability density function (PDF).
- In most of cases (in our class) will deal with **normal distribution**.
- We need to find appropriate mean and standard deviation to use normal distribution.

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\mu = \text{Mean}$$
$$\sigma = \text{Standard Deviation}$$

B0. Parameterizing the distribution using Maximum Likelihood



- Which distribution (left or right) is "likelihood" to represent the observed data?

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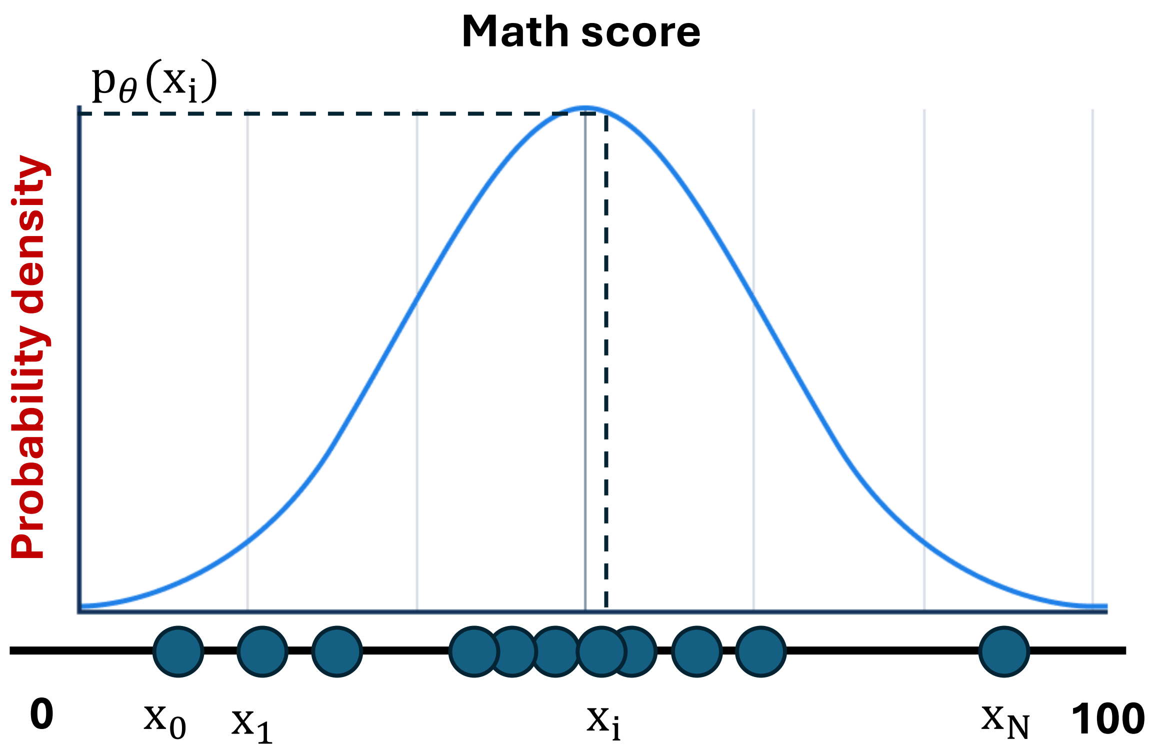
B0. Parameterizing the distribution using Maximum (log)Likelihood Estimation (MLE)



$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Likelihood function can be defined as:

$$L^*(\boldsymbol{\theta}) = \prod_{i=1}^{N} p_\theta(x_i)$$

$$\boldsymbol{\theta}^* = \operatorname{argmax} L^*(\boldsymbol{\theta})$$

- However, $L^*(\theta)$ could be too small because $p_\theta(x_i)$ are smaller than 1. Instead, we use:

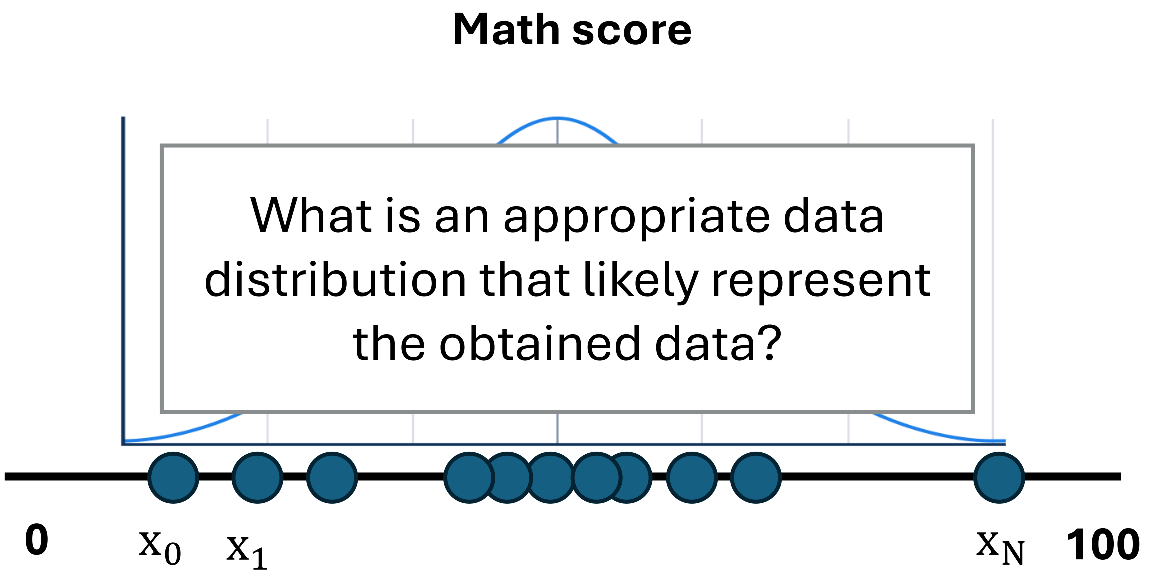
$$L(\boldsymbol{\theta}) = \log\left(\prod_{i=1}^{N} p_\theta(x_i)\right) = \sum \log(p_\theta(x_i))$$

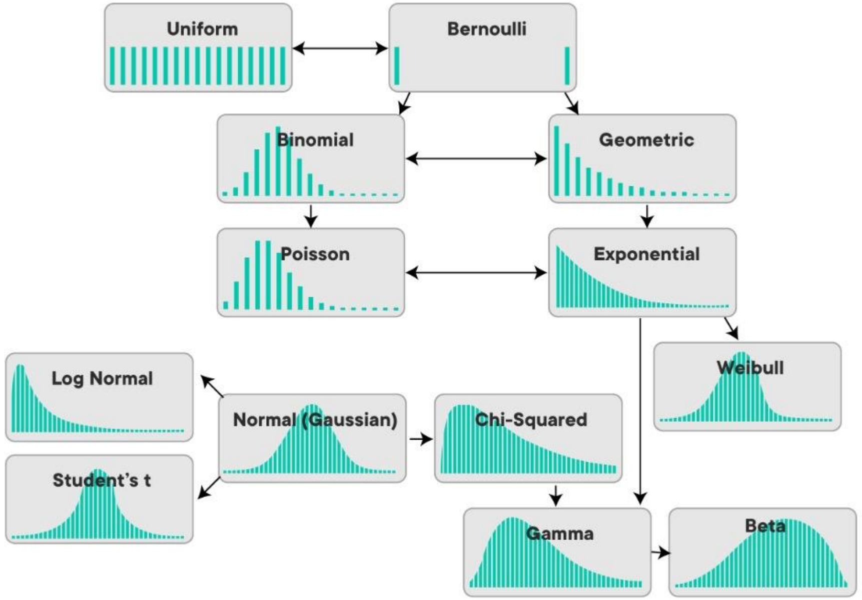
$$\boldsymbol{\theta}^* = \operatorname{argmax} L(\boldsymbol{\theta})$$

- Note that, $\log(f(x))$ and $f(x)$ is Monotonic Increasing.

Summary of Basics of Probability and Machine Learning

Key Takeaway: We can use MLE to parameterize obtained data distribution

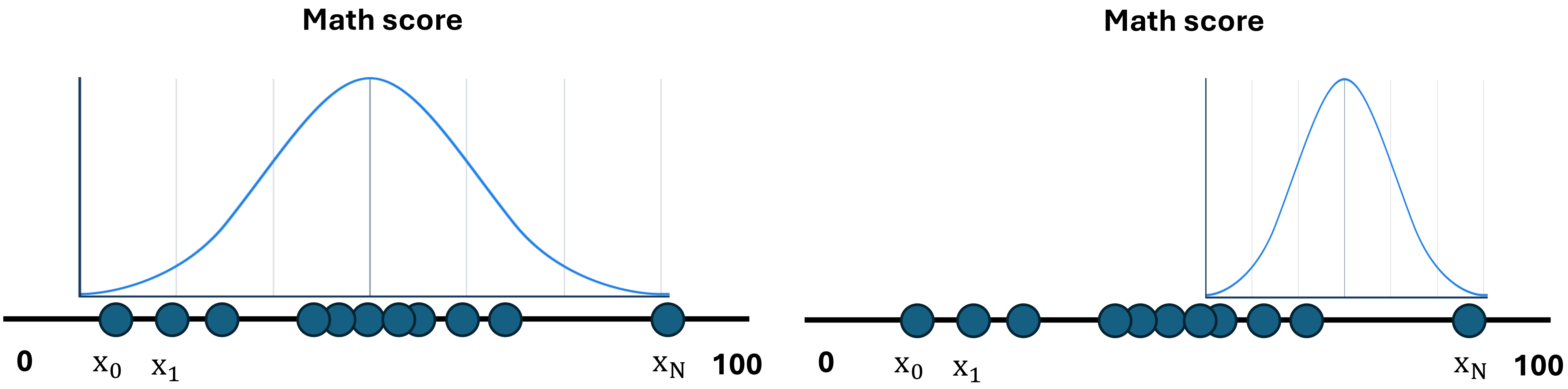




- Step1: Find appropriate probability density function (PDF) for obtained data distribution- we mostly use normal d istribution.

Summary of Basics of Probability and Machine Learning

Key Takeaway: We can use MLE to parameterize obtained data distribution



- Step2: Find appropriate parameter that fit chosen PDF to the data distribution.
  - This is done by Maximum Loglikelihood Estimation (MLE) method
  - In normal distribution we find mean and variance.

$$L(\boldsymbol{\theta}) = \sum \log(p_{\theta}(x_i))$$

$$\boldsymbol{\theta}^* = \operatorname{argmax} L(\boldsymbol{\theta})$$

Basics of Probability and Machine Learning

B1. Mathematical Expression of the ML model

- Imagine familiar equation: $\mathrm{y} = \mathrm{f}_\theta(\mathrm{x})$. This is a deterministic relationship between x and y. θ means that the equation is modelled with parameter θ. For example, we can imagine $\mathrm{y} = \theta\mathrm{x} + \mathrm{b}$

- However, in many cases of ML, x and y are NOT a single vector or scalar. Instead, they are “sampled” from a d ata distribution $\mathrm{p}_{\mathrm{data}}\ (x, \mathrm{y})$. In this case, we should model conditional distribution $\mathrm{p}(\mathrm{y}|x)$ and then sample y from it:

$$\mathrm{y} \sim \mathrm{p}_\theta(\mathrm{y}|x).$$

Imagine y as taste x as distribution of apples.
There are many red apples but less yellow, green apples.

B2. Bayes' rule

- In ML model that has $\mathrm{p}_{\theta}(\mathrm{y}|x)$ form, it is useful to know Bayes' rule:

$$p(y \mid x) = \frac{p(x, y)}{p(x)} = \frac{p(x \mid y)\, p(y)}{p(x)}$$

- $\mathrm{p}(\mathrm{y}|x)$: Given X, what is the probability of y
- $\mathrm{p}(x, \mathrm{y})$: Joint probability of x and y

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B3. Joint probability and it's marginalization

- **Problem definition: Driving a car from home to school.**
  - Variables could be {x: traffic level represented in average speed, y: time of day}.

Intuition: if we have 10 different conditions affecting the phenomena (time, speed, rain, health, etc). We can remove the effect of conditions by marginalizing them.

- $p(x = 10, y = 10)$ will have high value.
- $p(x = 2, y = 10)$ will have low value.

- **What should we do if we want to know $p(x)$ that doesn't care about time of day?**

$$p(x) = \int p(x, y) dy$$

B4. Marginalization of conditional probability

- **Problem definition: Driving a car from home to school in the rain.**
  - Variables could be $\{x$: amount of rain fall, y: departure time$\}$.
  - If I want to know about probability of being late, we can write down this by:

$$p(l|x, y)$$

  - Now, we want to know $p(l|y)$ which represents probability of being late conditioned on departure time.
    - This means, we now don't care about the rain. We want to know about **probability of being late** without being affected by rain fall.
  - We can account for all rain conditions by marginalizing the density function over x

$$p(l|y) = \int p(l|x, y)p(x|y)dx$$

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B5. Expectation of function under certain probability

- Assume $x \sim p(x)$. Then we can obtain expectation of function $f(x)$ as:

$$\mathbb{E}_{p(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\, p(\mathbf{x})\, d\mathbf{x}.$$

- In ML, this expression is used frequently when the model deals with the distribution – i.e.,$x \sim \mathrm{p}_{\mathrm{data}}(x)$ Expectation can be understood as the average of $f(x)$ under distribution $\mathrm{p}_{\mathrm{data}}(x)$.

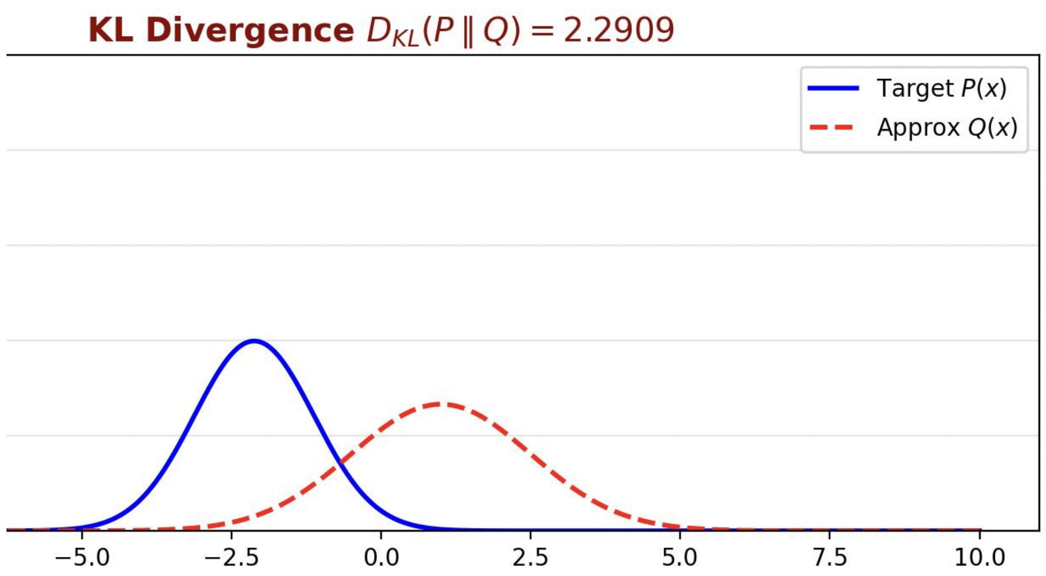
Example:
- We have 10 apples. 60% of apples are red, 10% of apples are yellow, and 30% of apples are green.
- Sweetness of red apples is 10, yellow apples have 8 sweetness, and green apples have 6.
- What is the expected sweetness?
  - 0.6*10 + 0.1*8 + 0.3*6 = 7.6

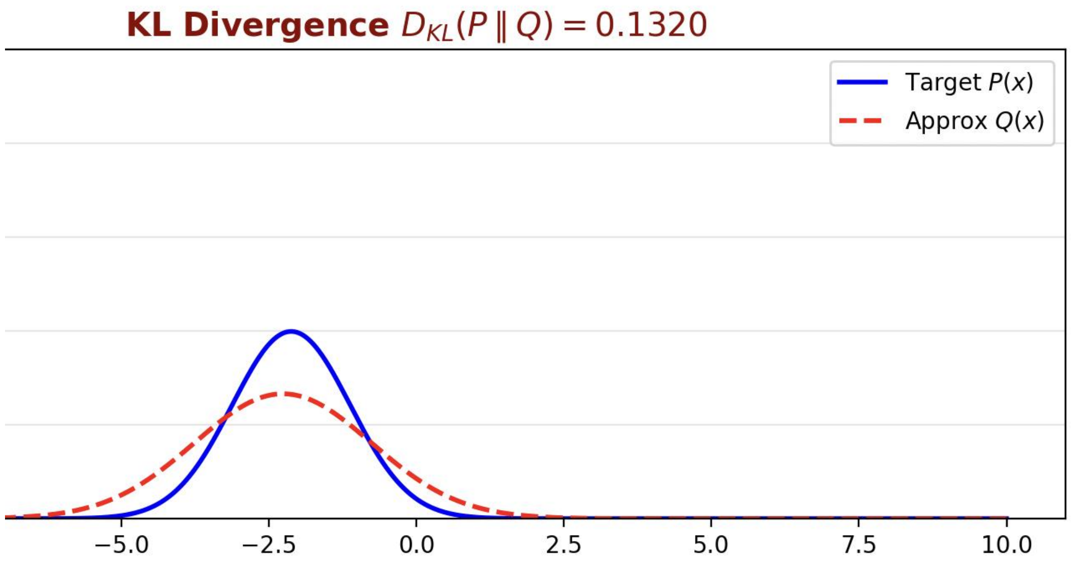
PR_code4.py

* $x \sim p(x)$ is usually called x is sampled from $\mathrm{p}(x)$.

B6. KL(Kullback-Leibler) divergence

- Assume we want to know the difference between two distribution. See two images below:





- We can measure the difference between two distributions by using KL divergence.

B6. KL(Kullback-Leibler) divergence

- Assume we have two probability density function $\mathrm{P}(\mathrm{x})$ and $\mathrm{Q}(\mathrm{x})$.

- Then, KL divergence $\mathrm{D}_{\mathrm{KL}}(\mathrm{P} \parallel \mathrm{Q})$ can be defined as

$$\sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

- This is equivalent to:

$$D_{\mathrm{KL}}(P \parallel Q) = \left(-\sum_{x \in \mathcal{X}} P(x) \log Q(x)\right) - \left(-\sum_{x \in \mathcal{X}} P(x) \log P(x)\right)$$

To understand KL divergence…
We should understand a concept of Entropy.

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B7. Entropy

- Assume we bought 5 lottery tickets. This lottery has 1% of change to get 1 billion won (10억), 50% chance to get 1000won, and 49% chance to get 0won.

Surprisingly, all the tickets gave me 1 billion won !!

The amount of surprise is related to the probability.

- To express our surprise in the equation, we can imagine

$$Q^* (p) = 1/p \text{ where, p is the probability}$$

- However, this could be extremely small in some cases. For the above case it will be:

$$Q^*(p) = 10^2 * 10^2 * 10^2 * 10^2 * 10^2$$

- To solve this issue, easy way is to use "log":

$$Q^{logcase}(p) = \log_{10} (1/p) = 2 + 2 + 2 + 2 + 2$$

We used $\log_{10}$ for easier computation, but we can use other such as 10 or 2 or e.

B7. Entropy

- Assume we bought 5 lottery tickets. T
  1000won, and 49% chance to get 0w

Example:

- We have 10 apples. 60% of apples are red, 10% of apples are yellow, and 30% of apples are green.
- Sweetness of red apples is 10, yellow apples have 8 sweetness, and green apples have 6.
- What is the expected sweetness?
  - $\boxed{0.6}*10 + \boxed{0.1}*8 + \boxed{0.3}*6 = 7.6$

- To solve this issue, easy way is to use “log”:

$$Q^{logcase}(p) = \log(1/p) = 2 + 2 + 2 + \log_{10}(2) + \log_{10}(100/49)$$

- Shouldn’t we also think about how likely it will happen (Expectation)?

$$Q(p) = \sum p * \log(1/p) = 0.01 * 2 + 0.01 * 2 + 0.01 * 2 + 0.5 * \log_{10}(2) + 0.49 * \log_{10}(100/49)$$

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B7. Entropy

- Assume we bought 5 lottery tickets.
  1000won, and 49% chance to get 0w

Example:

- We have 10 apples. 60% of apples are red, 10% of apples are yellow, and 30% of apples are green.
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- Interpreting the entropy:

$$Q_p(p) = \sum p * \log(1/p) = E_p[-\log(p(x))]$$

- In continuous phenomena we can express entropy as:

$$Q_p(p) = E_p(-\log(p)) = \int -p(x)\log(p(x))dx$$

Understanding KL divergence from the concept of entropy

Connecting Entropy with KL divergence

- Assume we bought 5 lottery tickets. This lottery has 1% of change to get 1 billion won (10억), 50% chance to get 1000won, and 49% chance to get 0won.

Now, 3 tickets got 1 billion won, 1 ticket got 1000 won, and 1 ticket got 0won.

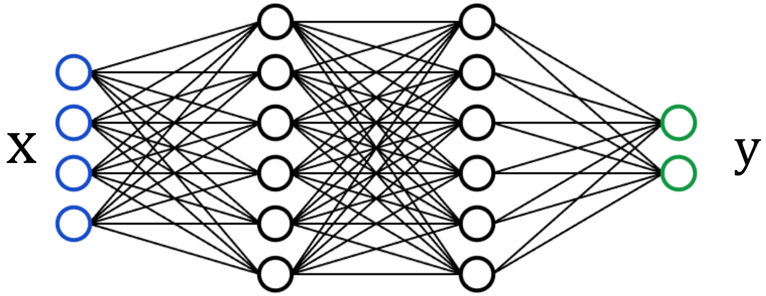
How about in this case?

- The fact was… that, due to the printing issue, the lottery had 60% of chance to get 1 billion won and 20% chance to get 1000 won, and 20% chance to get 0 won!!
- This will provide more surprise, so that the entropy could be larger.
- We use this increase of surprise as a difference of two distribution.

$$D_{\mathrm{KL}}(P \parallel Q) = \underbrace{\left(-\sum_{x\in\mathcal{X}} P(x)\,\log Q(x)\right)}_{\text{Entropy of mistake case}} - \underbrace{\left(-\sum_{x\in\mathcal{X}} P(x)\,\log P(x)\right)}_{\text{Entropy of original case}}$$

Overview on machine learning

- **Supervised learning**
  - Given $D = \{(x_i, y_i)\}$, find parameter θ that satisfies $y_i \approx f_\theta(x_i)$ or $f_\theta(y_i|x_i)$
  - Requires I.I.D (Independent, Identically Distributed) assumption.
    - Independent assumption: $(x_i, \mathrm{y}_i) \perp (x_\mathrm{j}, \mathrm{y}_\mathrm{j}) \forall \mathrm{i} \neq j$
    - Identically Distributed assumption: $(x_{train}, y_{train}), (x_{test}, y_{test}) \sim p(x, y)$
  - Ground truth assumption: training labels y_{train} are the ground-truth outputs.



- **Reinforcement Learning**
  - Train a policy using data collected through interaction with the environment.
  - I.I.D assumption is violated.
  - Ground truth assumption is also violated.
- **Unsupervised learning**
  - Given $D = \{x_i\}$, find parameter θ that represents hidden structure in the data.
  - I.I.D assumption: typically assumed.